Straight-Line Graphs

Quick reference sheet · GCSE Maths, Algebra · keep handy for revision

1. $y = mx + c$

m = gradient (steepness & direction) · c = y-intercept (where the line crosses the y-axis, when x = 0).

A bigger gradient is steeper. A negative gradient slopes downward left to right.

e.g. $y = 2x + 1 \rightarrow$ gradient **2**, y-intercept **1**.

2. Finding the gradient from two points

$m = (y_2 - y_1) \div (x_2 - x_1)$ — change in y over change in x.

(1, 3) and (4, 9) $\rightarrow$ $m = (9–3) \div (4–1) = 6 \div 3 =$ **2**

3. Plotting from a table, and parallel lines

Pick x-values, work out matching y-values, plot the points, then join with a straight line.

x	–1	0	1	2
y = 2x + 1	–1	1	3	5

Parallel lines have the same gradient (but a different y-intercept). e.g. $y = 2x + 1$ and $y = 2x – 3$ are parallel.

4. Finding the equation of a line

From a table: m = how much y changes each time x increases by 1. c = the y-value when x = 0.

x	0	1	2	3
y	4	7	10	13

y increases by 3 each time $\rightarrow$ m = 3. When x = 0, y = 4 $\rightarrow$ c = 4. $\rightarrow$ **$y = 3x + 4$**